Machine Learning for Signal Processing

Graduate Course – Short Handbook

3 Graduate Credits | **16 Instructional Weeks** | **3 Hours/Week**

Statistical inference → signal structure → learned inference → sequential inference → generative/diffusion models

1 Course Overview

Machine Learning for Signal Processing (MLSP) studies how machine learning can be integrated with the mathematical models, structures, and algorithms of signal processing.

Rather than presenting machine learning as a collection of independent algorithms, the course follows a central question:

What should be learned, and what should remain modeled?

The course develops a progression from classical statistical inference and signal representations to optimization-based reconstruction, learned inference algorithms, sequential estimation, and modern generative/diffusion models.

The central design principle is

Signal problem → Structure → Inference
→ Model-based algorithm → Learned component → Controlled evaluation

The emphasis is therefore on understanding, deriving, implementing, and comparing methods rather than simply using existing ML libraries.

2 Course Goals

By the end of the course, students should be able to:

- formulate signal-processing problems using statistical, Bayesian, and optimization-based inference;
- exploit low-rank, sparse, subspace, and learned representations;
- understand proximal optimization, Plug-and-Play methods, and algorithm unrolling for learned inference;
- connect state-space models and Kalman filtering with learned recurrence and state-space models;
- understand generative, score-based, and diffusion models as learned priors for inverse problems;
- conduct reproducible experiments with appropriate baselines, controlled comparisons, and failure analysis.

3 Target Students

The course targets graduate students and early-stage researchers in EE and related fields involving sensing, estimation, reconstruction, prediction, and structured data. It is especially suited to research in inverse problems, sparse recovery, sensor fusion, learned/model-based optimization, sequential estimation, learned dynamics, and generative reconstruction, with applications across communications, biomedical engineering, robotics, radar/sonar, control, and industrial monitoring.

4 Prerequisites

The prerequisite policy is primarily mathematical rather than algorithmic.

Required:

- Linear algebra: vectors, matrices, rank, eigenvalues/eigenvectors.
- Basic probability: random variables, expectation, variance, conditional probability, Gaussian distributions.
- Basic signals and systems.
- Multivariable calculus and the chain rule.
- Basic optimization and gradient descent.
- Basic Python programming.

Helpful but not required:

- SVD and numerical linear algebra.
- Fourier/transforms.
- Previous machine-learning or deep-learning exposure.
- Jupyter/scientific Python familiarity.

Previous ML/DL coursework is not required. The necessary ML and neural-network foundations are introduced before they are used in later learned algorithms.

5 Tree-like Course Progression

The course is organized as a dependency tree rather than a flat collection of machine-learning topics, as shown in Figure 1.

The conceptual progression is

Inference → Representation → Optimization
→ Learned Algorithms → Sequential Inference → Generative/Diffusion Inference

Each module answers a question left open by the previous one.

6 Concise Module Timetable

The following table shows only representative topics.

Weeks	Module	Representative topics
1–3	**I. Statistical Learning, Inference & Look-Ahead**	ML/MAP/MMSE; detection and regression; neural-network foundations; GMM/EM
4–6	**II. Representation & Signal Structure**	SVD/PCA; low-rank models; sparsity and compressed sensing; autoencoders
7–10	**III. Learning the Inference Algorithm**	Inverse problems; proximal optimization; ISTA/FISTA; Plug-and-Play; algorithm unrolling
11–13	**IV. Structured & Sequential Inference**	State-space models; Kalman filtering; nonlinear filtering; learned dynamics
14–16	**V. Generative & Diffusion Models for Inverse Problems**	Generative priors; score-based models; diffusion; posterior sampling and reconstruction

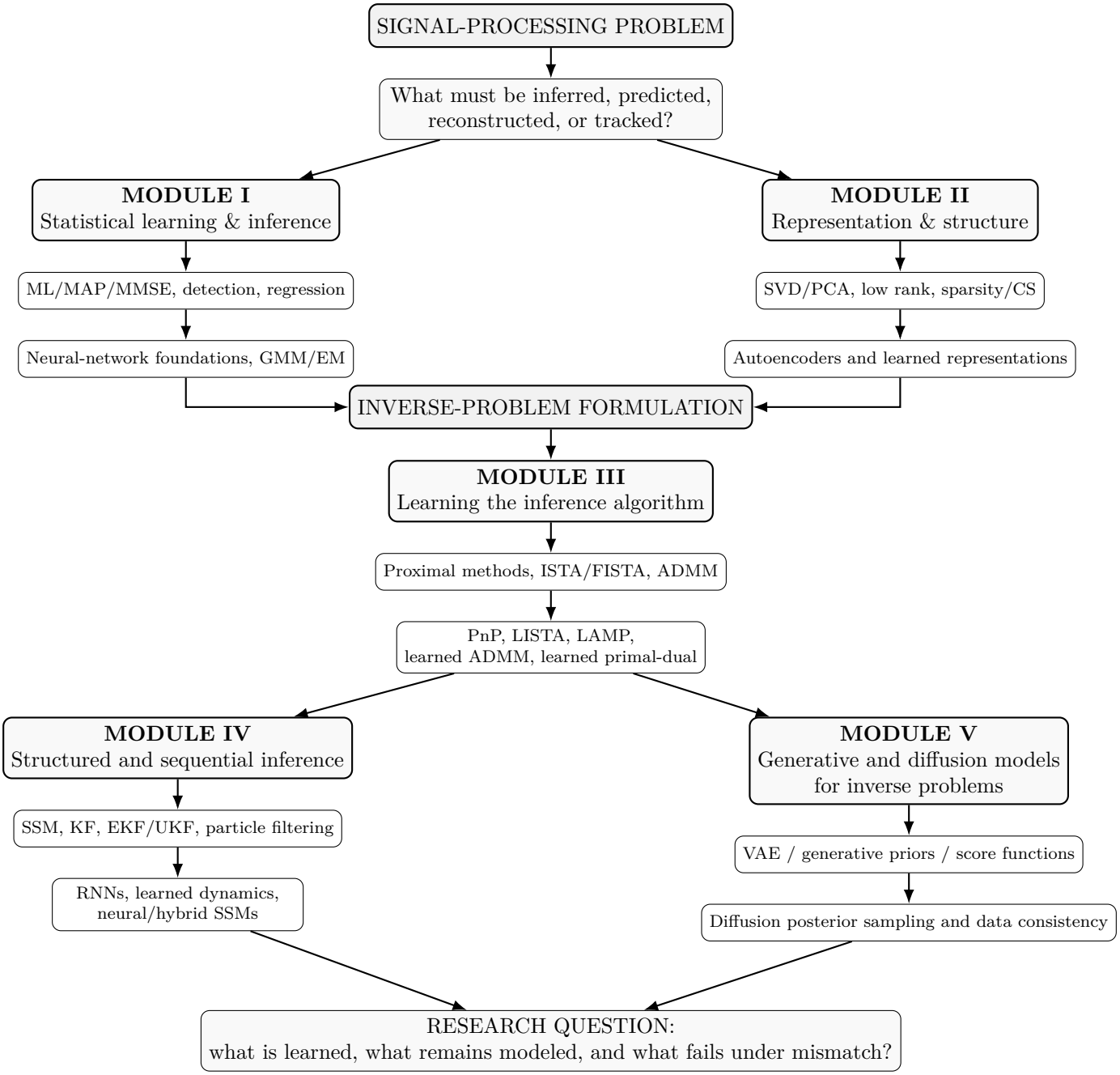

Figure 1: Five-module MLSP dependency map.

7 Weekly Instructional Structure

Each week consists of approximately

2 h Lecture + 1 h TA / Recitation / Lab

Lectures focus on mathematical foundations, derivations, conceptual connections, signal-processing examples, and selected research papers.

TA sessions focus on implementation, numerical experiments, visualization, debugging, homework support, and selected extensions.

Python is the primary implementation environment, with NumPy, SciPy, Matplotlib, scikit-learn, and PyTorch used as appropriate.

8 Assessment

Assessment is organized around four abilities:

1. **Understand** – explain the mathematical and signal-processing foundations.
2. **Derive** – connect assumptions, objectives, and algorithms.
3. **Implement** – reproduce and experiment with representative methods.
4. **Analyze** – compare methods and identify assumptions, limitations, and failure modes.

The course has:

Assessment component	**Number**
Homework sets	4
Final project	1
Final examination	1

The four homework sets are designed to follow the major stages of the course progression, while the final project provides an opportunity to integrate modeling, inference, implementation, and evaluation.

A typical project compares a classical/model-based baseline with a learned or hybrid method:

$$\text{Model-based baseline} \quad \longleftrightarrow \quad \text{Learned / hybrid method.}$$

The objective is not simply to obtain a lower numerical error, but to understand *why* the methods behave differently and under what conditions the learned component provides an advantage.

9 The MLSP Perspective

Throughout the course, students repeatedly analyze four components:

$$\boxed{\text{Measurement Model} + \text{Signal Structure} + \text{Inference Algorithm} + \text{Learned Component}}$$

For example,

$$\text{Sparse signal} \rightarrow \ell_1\text{-regularized inverse problem} \rightarrow \text{ISTA} \rightarrow \text{LISTA},$$

and later,

$$\text{State-space model} \rightarrow \text{Kalman filtering} \rightarrow \text{learned dynamics},$$

and finally,

$$\text{Measurement model} + \text{Generative prior} \rightarrow \text{posterior inference} \rightarrow \text{diffusion-based reconstruction}.$$

Thus, the course asks not only

"What ML algorithm should I use?"

but more fundamentally

"What structure do I know, what can I learn, and how should the two be combined?"